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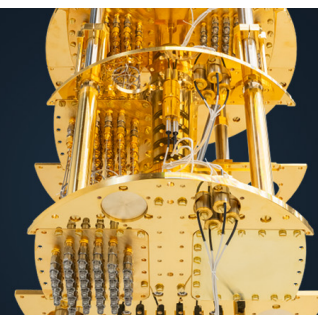
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ABSTRACT

In the industrial semiconductor plasma processing, plasma density and its spatial distribution are critical for the understanding of discharge physics and the enhancement of the processing yield via monitoring of the status of plasma processing. Thus, real-time monitoring of plasma density and uniformity in the wafer region during wafer processing have been receiving significant attention. In this study, we developed an electrode-embedded flat-cutoff array sensor for the real-time measurement of electron density uniformity during wafer processing. We measured the spatial distribution of electron density on the wafer surface, compared it with the results measured in the bulk plasma region, and found different plasma distributions between the two where this difference can be attributed to local electron kinetics. We also performed plasma etching of amorphous carbon materials, which showed a clear correlation with the electron density distribution measured by the flat-cutoff array sensor. This indicates that the electron density distribution on the wafer surface is crucial for the process result. Thus, the array sensor developed in this study can be applied not only to measure plasma uniformity but also to predict the wafer process beforehand, making it highly significant for industrial plasma processing methods, such as virtual metrology.

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The real-time diagnostics of plasma parameters have garnered widespread attention in a semiconductor processing field for the important role they play in real-time abnormality detection and feedback control during semiconductor processing.^{1–6} Among all the plasma parameters that can affect its processing, the most crucial factors involve electron density and its uniformity, as this influences the number of radical species and ions that contribute to etching or deposition.^{1,7} In addition, the spatial distribution of electrons is closely related to the spatial distribution of plasma potential, radicals, and ions.^{2,8} In other words, electron density and its uniformity affect the processing throughput, quality, and uniformity of the resulting plasma.^{7,8}

Conventional real-time process monitoring techniques in the semiconductor industry include end point detectors that use optical emission spectrometry (OES) and electrical sensors. However, it is difficult to measure plasma parameters with high precision using OES or

electrical sensors as they are indirect methods for plasma diagnostics. Moreover, these conventional monitoring methods face challenges in measuring plasma spatial uniformity, which affects the uniformity of the process results in the 300 mm wafer scale as well as in large-area plasmas. Therefore, it is necessary to develop a direct diagnostic method for industrial plasma processing that can accurately measure the uniformity of processed plasmas in real time.

Studies have been conducted on various direct methods for measuring plasma density and uniformity, such as electrostatic,⁹ optical,⁷ and microwave^{10–13} diagnostic methods. The electrostatic method, represented by the Langmuir probe, can be used to measure several plasma parameters, including the ion density, electron density, electron temperature, plasma potential, floating potential, and electron energy distribution function. However, it is rarely used for the monitoring of plasma processing as the probe tip is easily deformed by the

processing gas. Moreover, the plasma can be perturbed by the insertion of the probe.¹⁴ Optical diagnostics, such as OES, can take measurements outside the processing chamber, which allows for process parameters to be measured in real-time without perturbing the plasma.^{15,16} In this case, the spatial distribution of plasma can be measured by applying an additional optical system;¹⁶ however, it is difficult to quantify plasma parameters given that absolute calibration requires a significant effort. Optical diagnostic methods are also vulnerable to window contamination by the deposition layers during processing. Furthermore, their measurements of spatial distribution are not adequate because the measured signal is a line integrated along the line of sight.¹⁶

Finally, the microwave diagnostic method can also be used to measure plasma parameters. Even though the process gas deforms the probe tip, microwaves can still penetrate the dielectric film. Thus, the microwave probe can be regarded as a plasma diagnostic method applicable to etching or deposition processing.¹⁰ Many microwave diagnostic methods exist, including cutoff,¹⁰ plasma absorption,¹¹ multipole resonance,¹² and hairpin probes.¹³ However, because this is an invasive method in which the probe must be inside the plasma,¹⁴ it can lead to plasma perturbation and chamber contamination. This is not allowed in an industrial mass production setting. Furthermore, measurement of plasma parameters on the wafer surface is limited owing to the physical limitations of the probe structure. In summary, there is a huge demand for a noninvasive sensor that can, in real-time, measure plasma parameters with high precision during industrial processing.

Typical tools used in noninvasive microwave diagnostic methods for measuring electron density are the flat-cutoff,^{17–21} curling,²² and

planar multipole resonance sensors.²³ A flat-cutoff sensor can measure the electron density at the plasma-sheath boundary in real time, even if a dielectric, conductive, or patterned wafer is placed on the sensor.¹⁹ However, there are limited studies on measuring the spatial distribution at the plasma-sheath boundary, even though the spatial distribution of plasma parameters in this region is closely related to the processing results.^{1,7} Moreover, despite the importance of measuring plasma parameters at the wafer surface during processing, the real-time measurement of the plasma uniformity has rarely been attempted using a sensor embedded in the electrode on which the wafer is placed.

For this study, we developed an electrode-embedded-type flat-cutoff array sensor with a switching circuit. We performed real-time measurements of the spatial distribution of electron density during the amorphous carbon removal process, using an Ar/O₂ gas mixture plasma. By comparing the spatial distribution of electron density measured on the wafer surface with that in the bulk plasma, we confirmed that there was a difference between the two, which was caused by the local electron kinetics. We measured the plasma density distribution during the removal of amorphous carbon and placed the amorphous carbon/Si wafer on the flat-cutoff array sensor. We observed the etch rate of the amorphous carbon and the spatial distribution of its electron density, which showed a clear correlation with the electron density distribution as measured by the electrode-embedded flat-cutoff array sensor.

Figure 1(a) shows a schematic of the experimental setup used, including an inductively coupled plasma (ICP) chamber. The ICP chamber had a cylindrical shape with an inner radius of 700 mm, a height of 450 mm, and a dielectric window-to-electrode distance of

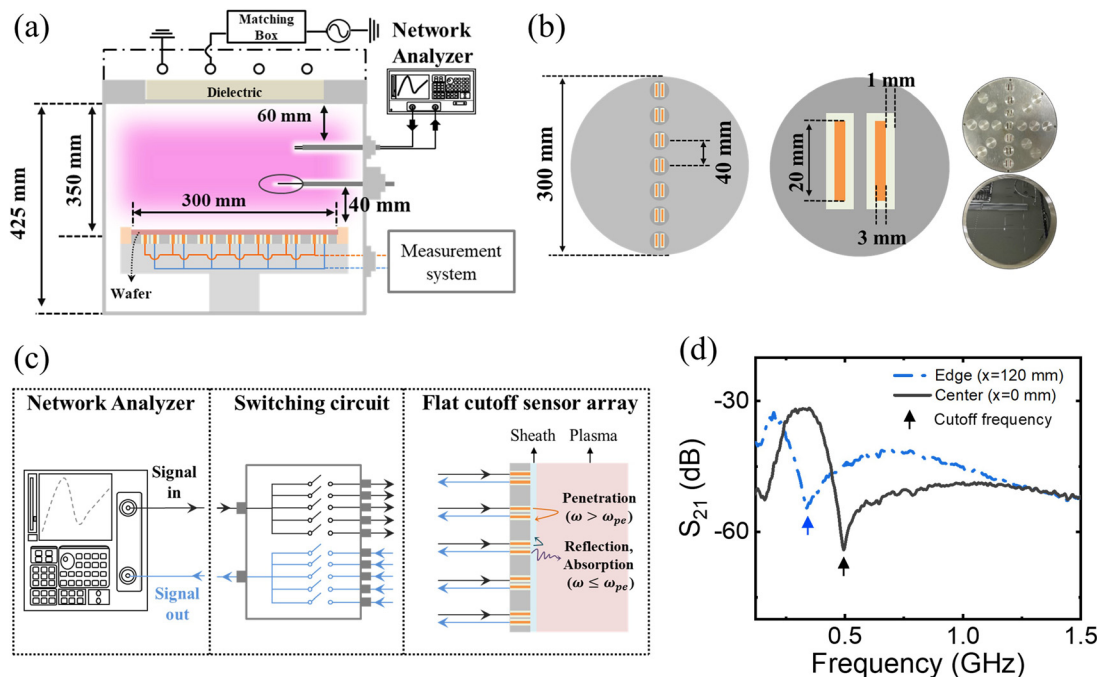


FIG. 1. (a) Schematic of the experimental setup, (b) flat-cutoff array sensor, (c) measurement system for the developed sensor, and (d) transmission spectrum (S_{21}) of the electrode-embedded type flat-cutoff sensor as measured at the center (black solid line) and edge (blue dotted line) of the electrode for a pressure of 100 mTorr and an input power of 1500 W in Ar/O₂ discharge.

350 mm. We connected a 13.56 MHz power supply to a double-stacked planar-type antenna through an L-type auto-matching network. An input power was maintained steadily at 1500 W during the experiment. The supply of argon and oxygen gases was controlled by a mass flow controller at a fixed flow rate of 50 sccm for each. We used roots and turbo-molecular pumps to maintain a base pressure below 1×10^{-6} Torr, as measured by a cold cathode gauge. The experimental pressure was 100 mTorr, which we measured using a capacitive manometer gauge calibrated with a standard pressure gauge at the Korea Research Institute of Standards and Science. We etched a wafer on a stack of an amorphous carbon (2000 nm)/Si substrate. To measure the etch rate of the wafer, we performed field-emission scanning electron microscopy (FE-SEM, Hitachi S-4800). A flat-cutoff array sensor and an invasive cutoff probe were used to measure the electron density on the wafer surface and in the bulk plasma region, respectively. An rf-compensated single Langmuir probe made of tungsten wire with a length of 10 mm and a diameter of 0.1 mm was placed 40 mm above the wafer. The probe system contained a floating loop reference probe, and resonance filters (13.56 MHz and 27.12 MHz) were used to reduce the rf distortion of the I–V characteristics.⁹

We placed an invasive-type cutoff probe 60 mm below the dielectric window and embedded a flat-cutoff array sensor in the electrode. Figure 1(b) shows the flat-cutoff sensor array used in the experiment. It is comprised of a radiating and detecting antenna, an insulator, and a ground plate. The distance between the radiating and detecting antennas, thickness of the antenna, length of the antenna, and thickness of the insulator were 8, 3, 20, and 1 mm, respectively. The flat-cutoff sensor used consisted of seven sensors placed in a row, with a distance of 40 mm between the adjacent probes.

Figure 1(c) shows the flat-cutoff array sensor measurement system, which consists of three parts: (i) a network analyzer, (ii) a switching circuits, and (iii) a flat-cutoff array sensor. The network analyzer (National Instruments, PXIe-5630) had two ports: one for emitting an electromagnetic (EM) wave and the other for detecting the EM wave passing through the measurement system and the plasma. We programmed the switching circuit (National Instruments, PXI-2596) to automatically distribute the EM wave emitted from the network analyzer to the sensor array so that the spatial distribution of electron density could be measured. In addition, we measured the transmittance of the EM wave (S_{21}) from each probe to the network analyzer through a switching circuit. We connected the switching circuit between the network analyzer and the flat-cutoff array sensor via a coaxial cable (50 Ω) to obtain an S_{21} spectrum for each flat-cutoff sensor.

The flat-cutoff sensor measured the electron density through the EM wave transmission characteristics of the plasma. Equation (1) determines the dispersion relation of EM waves propagating in plasma without a direct current (DC) magnetic field that of an ordinary wave in a DC magnetic field

$$\omega^2 = \omega_{pe}^2 + c^2 k^2. \quad (1)$$

In Eq. (1), ω is the EM wave frequency, ω_{pe} is the electron plasma frequency, k is the wavenumber, and c is the speed of light. According to Eq. (1), the plasma reflects the EM wave when $\omega \leq \omega_{pe}$, and the EM wave can penetrate the plasma when $\omega > \omega_{pe}$. Therefore, the minimum peak of the wave transmission spectrum indicates the electron plasma frequency. Similar studies have demonstrated that, using plasma oscillation probes, the minimum peak of the transmission spectrum (cutoff frequency) corresponds to the electron plasma frequency.¹⁰ Hence, by measuring the cutoff frequency, the electron density can be obtained using the following equation:

$$n_e = \frac{\omega_{pe}^2 \epsilon_0 m_e}{e^2}. \quad (2)$$

In Eq. (2), n_e is the electron density, ϵ_0 is the vacuum permittivity, m_e is the electron mass, and e is the elementary charge. The measurement uncertainty of the flat-cutoff sensor was less than $\pm 2\%$.¹⁹

Figure 1(d) shows the S_{21} spectra measured at the center and edge of the electrode-embedded-type flat-cutoff array sensor, respectively, for an input power of 1500 W, a gas pressure of 100 mTorr, and Ar and O₂ gas flow rates of 50 sccm. Both curves obtained with the flat-cutoff sensors in Fig. 1(d) detect and quantify a clear *N-shaped* signal and the minimum transmission peak. Moreover, the electron density at the central region is higher than that at the edge, which is expected given that the cutoff frequency is proportional to the square root of the electron density in Eq. (2).

Figure 2 shows the radial profiles of the electron density at both the plasma bulk and wafer surface, the electron temperature (T_e), the reaction rate of the O radical generation, and the etch rate of the amorphous carbon film. Electron density measurements were performed via the flat-cutoff array sensor using an amorphous carbon/Si wafer placed on the sensor. The experimental conditions are identical to those shown in Fig. 1(d). We calculated the non-uniformity of the spatial distribution to compare the electron density and etch rate spatial distributions. The non-uniformity of the electron density and amorphous carbon film etch rate was calculated as follows: $U = (x_{\max} - x_{\min})/2x_{\text{average}}$, where $x_{\max}$, $x_{\min}$, and x_{average} denote the

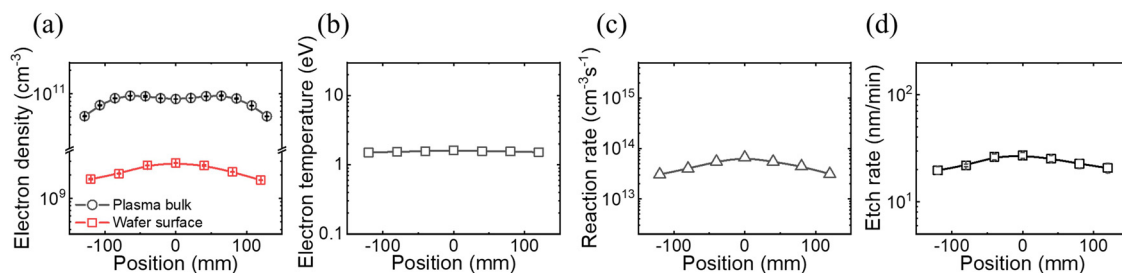


FIG. 2. Spatial distribution of the (a) electron density of plasma bulk and wafer surface, (b) electron temperature 40 mm above the wafer surface, (c) reaction rate of the O radical generation, and (d) etch rate of ACL at a fixed gas pressure of 100 mTorr and an input power of 1500 W by Ar/O₂ discharge.

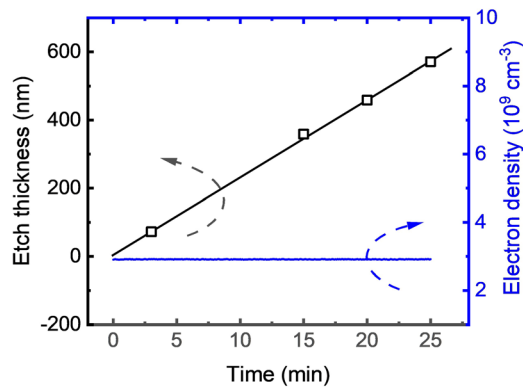


FIG. 3. Electron density (blue solid line) and etch thickness of the ACL film (black symbol), measured over time at a fixed gas pressure of 100 mTorr and an input power of 1500 W by Ar/O₂ discharge.

maximum, minimum, and average values measured, respectively. From Fig. 3(a), we can see that at the plasma bulk, the maximum value of n_e was $9.43 \times 10^{10} \text{ cm}^{-3}$, the minimum value was $5.15 \times 10^{10} \text{ cm}^{-3}$, and the non-uniformity was 28.46%; whereas, at the wafer surface, the maximum value of n_e was $2.84 \times 10^9 \text{ cm}^{-3}$, the minimum value was $1.77 \times 10^9 \text{ cm}^{-3}$, and the non-uniformity was 22.88%. The error bars in Fig. 2(a) represent the measurement uncertainty of the electron density. In our experiment, the measurement uncertainty of the invasive-type cutoff probe²⁴ was less than $\pm 1.534\%$, while the measurement uncertainty of the developed flat-cutoff sensor¹⁹ was less than $\pm 1.913\%$. The frequency resolution of the vector network analyzer was 1.87 MHz, and the electron density resolution calculated from the frequency resolution was $4.34 \times 10^4 \text{ cm}^{-3}$.

The electron density profiles measured in the plasma bulk show a saddle shape in the radial direction and a convex shape on the wafer surface. The radial distribution near the antenna can be understood using the local electron kinetics. When the electron energy relaxation length (λ_e) is longer than the dimensions of the chamber (non-local electron kinetics), the heated electrons generated around the antenna can reach any region inside the chamber, which can then have a center-peak distribution. However, when λ_e is sufficiently short in comparison to the chamber dimensions (local electron kinetics), the region where electron heating occurs can be limited to the ionization region. The transition from non-local to local kinetics depends on external parameters, such as gas pressure²⁵ and the addition of molecular gas.²⁶ Here, λ_e can be calculated as: $\lambda_e = (\lambda_m \lambda_{ine}/3)^{0.5}$, where λ_m and λ_{ine} are the electron mean free paths for momentum transfer and inelastic collisions, respectively. In our experimental conditions, λ_e is less than 50 mm at electron energies below 30 eV, which is much shorter than the chamber radius and height. In the local electron kinetics regime, ionization by energetic electrons mainly occurs near the antenna, where the maximum electrostatic or electromagnetic field and radial electron density profiles follow the antenna shape.²⁷ By contrast, the radial distribution of the electron density measured by the flat-cutoff array sensor on the wafer surface has a convex shape. The radial profile of the electron density relaxes toward the wafer surface by electron diffusion.²⁷ Because there is a sufficient diffusion of electrons that are generated in the plasma bulk to the electrode surface, the electron density distribution at the electrode surface can be changed from a saddle shape to a center-peak shape.

On the other hand, the spatial distribution of the electron temperature has a flat shape with a maximum value of 1.63 eV, a minimum value of 1.51 eV, and a non-uniformity of 3.74%, as shown in Fig. 2(b). Therefore, the spatial distribution of O radicals, which is a function of the electron density and electron temperature, is expected to be strongly influenced by the electron density, which presents a convex shape distribution rather than the flat shape of the electron temperature distribution. In addition, because O radicals are known to be the main etchant in amorphous carbon etching,²⁸ the spatial distribution of the ACL etch rate is expected to be similar to that of the electron density profile. To estimate the spatial distribution of O radicals, we calculated the reaction rate of the O radical generation based on the measured plasma parameters. The reaction rate of the O radical generation by electron-impact collisions with oxygen molecules can be described as $R = n_e n_{O_2} K_{diss}$, where R is the reaction rate, n_{O_2} is the O₂ gas density, and K_{diss} is the rate constant of the dissociation, given by $k_{diss} = 6.86 \times 10^{-9} e^{-\frac{6.22}{T_e}}$.^{5,29} Figure 2(c) shows the spatial distribution of the reaction rate of O radical generation. Similar to the electron density spatial distribution, the reaction rate spatial distribution also presents a convex shape, with a non-uniformity of 38.94%. Because the spatial distribution of T_e is more uniform than that of the spatial distribution of n_e , the non-uniformity of the reaction rate is more dominant than the electron density under our experimental conditions. In Fig. 2(d), the maximum value of the amorphous carbon film etch rate is 27.2 nm/min, the minimum value is 19.7 nm/min, and the non-uniformity is 16.13%. As expected, the spatial distribution of the etch rate of the amorphous carbon film shows a shape similar to that of the electron density distribution and reaction rate at the wafer surface, in contrast to that at the plasma bulk and of the electron temperature. Consequently, the spatial distributions of the electron density at the wafer surface and the amorphous carbon etch rate are in good agreement. This implies that real-time electron density uniformity measurements, made using the developed sensor herein reported, can be a useful indicator in predicting and controlling the uniformity of the process results. The electron density spatial distribution measured with the flat-cutoff sensor is in good agreement with that of the etch rate spatial distribution, albeit with a slight non-uniformity difference of 6.75 percentage point. The ICP discharge, because the sheath potential is low, comes down to the effect of radicals acting as a significant factor in etching,^{1,7} instead of ions. Conversely, in an ion-dominant etching environment, where a high sheath voltage is applied, such as when using a capacitively coupled plasma or ICP-biased plasma,¹ we expect that the discrepancy between the electron density and process uniformity would be further reduced.^{1,16}

Figure 3 shows the measured electron density and etch thickness of the ACL depending on the processing time. Electron density measurements, taken every 5 s, and ACL etching were performed at the center of the electrode. During the process, for up to 25 min, the mean electron density was measured at $2.872 \times 10^9 \text{ cm}^{-3}$, with a standard deviation of $5.278 \times 10^6 \text{ cm}^{-3}$. The thicknesses of the ACLs etched for 3, 10, 15, and 20 min were 72.6, 358.9, 459.0, and 571.0 nm, respectively; and the ACL-etched thickness increased linearly with time. From this observation, it is expected that the prediction of the etch thickness using real-time electron density measurements is possible.

In conclusion, we developed an electrode-embedded flat-cutoff array sensor with a switching circuit and performed real-time electron density and uniformity measurements during the removal process of

plasmas. To confirm the reliability of this technique, we compared the spatial distribution of electron density at the wafer surface with that at the plasma bulk and found that both sets of data were in good agreement. Through our study, we demonstrated that the use of a reliable real-time plasma monitoring technique for plasma processing is made possible by using flat-cutoff array sensors, which can contribute to improving the uniformity of the plasma process reactors.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

H. J. Yeom: Data curation (equal); Formal analysis (equal); Investigation (equal); Visualization (equal); Writing – original draft (equal). **Min Young Yoon:** Formal analysis (equal); Investigation (equal). **Gwang-Seok Chae:** Formal analysis (equal); Investigation (equal). **Jung-Hyung Kim:** Conceptualization (equal); Data curation (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal). **Shinjae You:** Formal analysis (equal); Validation (equal). **Hyo-Chang Lee:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Project administration (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal).

DATA AVAILABILITY

Data supporting the findings of this study are available from the corresponding authors upon reasonable request.

REFERENCES

- ¹H.-C. Lee, *Appl. Phys. Rev.* **5**, 011108 (2018).
- ²B. Wu, A. Kumar, and S. Pamarthi, *J. Appl. Phys.* **108**, 051101 (2010).
- ³V. M. Donnelly and A. Kornblit, *J. Vac. Sci. Technol. A* **31**, 050825 (2013).
- ⁴T. Lill and O. Joubert, *Science* **319**, 1050 (2008).
- ⁵M. A. Lieberman and A. J. Lichtenberg, *Principles of Plasma Discharges and Materials Processing* (John Wiley & Sons, Inc., Hoboken, NJ, 2005).
- ⁶F. F. Chen, *Introduction to Plasma Physics and Controlled Fusion* (Springer International Publishing, Cham, 2016).
- ⁷S. Yun and G. R. Tynan, *J. Appl. Phys.* **89**, 911 (2001).
- ⁸J. Schulze, Z. Donkó, E. Schüngel, and U. Czarnetzki, *Plasma Sources Sci. Technol.* **20**, 045007 (2011).
- ⁹V. A. Godyak, R. B. Piejak, and B. M. Alexandrovich, *Plasma Sources Sci. Technol.* **1**, 36 (1992).
- ¹⁰J. H. Kim, D. J. Seong, J. Y. Lim, and K. H. Chung, *Appl. Phys. Lett.* **83**, 4725 (2003).
- ¹¹H. Kokura, K. Nakamura, I. P. Ghanashev, and H. Sugai, *Jpn. J. Appl. Phys.* **38**, 5262 (1999).
- ¹²M. Lapke, T. Mussenbrock, and R. P. Brinkmann, *Appl. Phys. Lett.* **93**, 051502 (2008).
- ¹³R. B. Piejak, V. A. Godyak, R. Garner, B. M. Alexandrovich, and N. Sternberg, *J. Appl. Phys.* **95**, 3785 (2004).
- ¹⁴V. Godyak, *Phys. Plasmas* **24**, 060702 (2017).
- ¹⁵J. Li, F.-X. Liu, X.-M. Zhu, and Y.-K. Pu, *J. Phys. D: Appl. Phys.* **44**, 292001 (2011).
- ¹⁶I. J. Kim and I. Yun, *IEEE Trans. Ind. Electron.* **63**, 5674 (2016).
- ¹⁷H. C. Lee, J. H. Kim, and H. J. Yeom, “Device for measuring plasma ion density and apparatus for plasma diagnostics using the same,” Korea patent 102340564, US patent 17222937, EU patent EP21167098.9, China patent 2021110374496.9, Japan patent 202162613 (2021).
- ¹⁸D. W. Kim, S. J. You, S. J. Kim, J. H. Kim, J. Y. Lee, W. S. Kang, and M. Hur, *Plasma Sources Sci. Technol.* **28**, 015004 (2019).
- ¹⁹H. J. Yeom, J. H. Kim, D. H. Choi, E. S. Choi, M. Y. Yoon, D. J. Seong, S. J. You, and H.-C. Lee, *Plasma Sources Sci. Technol.* **29**, 035016 (2020).
- ²⁰H. J. Yeom, K. H. You, J.-H. Kim, and H.-C. Lee, *Plasma Sources Sci. Technol.* **30**, 065012 (2021).
- ²¹H. C. Lee, J. H. Kim, D. J. Seong, and H. J. Yeom, “Planar type plasma diagnosis apparatus, wafer-type plasma diagnosis apparatus in which planar-type plasma diagnosis apparatus is buried, and electrostatic chuck in which planar-type plasma diagnosis apparatus is buried,” Korea patent 1020200095022, US patent 17050373, EU patent 19912976.8, China patent 201980028803.9, Japan patent 2021514264 (2020).
- ²²A. Pandey, W. Sakakibara, H. Matsuoka, K. Nakamura, and H. Sugai, *Appl. Phys. Lett.* **104**, 024111 (2014).
- ²³C. Schulz, T. Styrnoll, P. Awakowicz, and I. Rolfes, *IEEE Trans. Instrum. Meas.* **64**, 857 (2015).
- ²⁴J.-H. Kim, K.-H. Chung, and Y.-H. Shin, *Metrologia* **42**, 110 (2005).
- ²⁵L. D. Tsendin and U. Fiz, *Usp. Fiz. Nauk* **180**, 139 (2010).
- ²⁶H. C. Lee, H.-J. Hwang, Y.-C. Kim, J. Y. Kim, D.-H. Kim, and C.-W. Chung, *Phys. Plasmas* **20**, 033504 (2013); H. C. Lee, M. H. Lee, and C. W. Chung, *Appl. Phys. Lett.* **96**, 041503 (2010).
- ²⁷J. H. Kim, Y. C. Kim, and C. W. Chung, *Phys. Plasmas* **22**, 073502 (2015).
- ²⁸D. Choi, H. J. Yeom, K. H. You, J. H. Kim, D. J. Seong, E. Yoon, and H.-C. Lee, *Carbon N.Y.* **162**, 423 (2020).
- ²⁹J. T. Gudmundsson, *J. Phys. D: Appl. Phys.* **35**, 328 (2002).